MO7019: Methods for Climate Science
Statistika Metoder For Klimatvetenskap (7.5 hp)
Exercises

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Further exercise: Significance via non-parametric and Monte-Carlo methods

This is a practical exercise to check whether two time series have the same mean, and you need to use your computer.
Consider the monthly NAO time series (1950-2024) in
https://psl.noaa.gov/data/correlation/nao.data

Download the data and split it into two time series x_t and y_t, $t = 1, \ldots 450$.

1. Compute the means $\overline{x}$ and $\overline{y}$, and let $t_o = \overline{y} - \overline{x}$.
2. Let $N = 1000$, and $p = 150$, and $T = array(1, 1 : N)$,
   for i=1:N
      draw randomly subsamples from x and y of size p,
      compute their means m_1 and m_2, and let $T(i) = m_2 - m_1$
   end
3. Draw a histogram of T, and locate t_o on it. Conclude
4. Calculate the approximate p-value
5. Use the Kolmogorov-Smirnov test to check if the two time series come from the same probability distribution.

Compute the p-value with the KS test and compare it with the above p-value of question 4. Comment